

Corollaries of the Axioms of Probability

Theorem

Given events E and F (in $\mathcal{F}$), if $E \subset F$
then $P(E) \leq P(F)$

Proof:

Since $E \subset F$, can partition F into
two disjoint sets: disjoint (mutually exclusive)

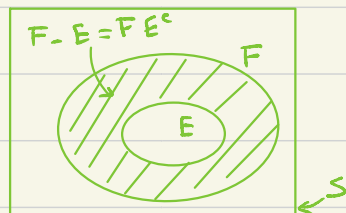
$$F = E \cup (F-E)$$

Hence by Axiom 3 (its corollary), have

$$P(F) = P(E) + P(F-E) \quad \leftarrow \text{eq. ④ of last lecture}$$

≥ 0 by Axiom 1

$$\therefore P(F) \geq P(E) \quad \square$$



Corollary: If $E \subset F$, then

$$P(F-E) = P(F) - P(E)$$

Theorem

For any events E and F (in $\mathcal{F}$),

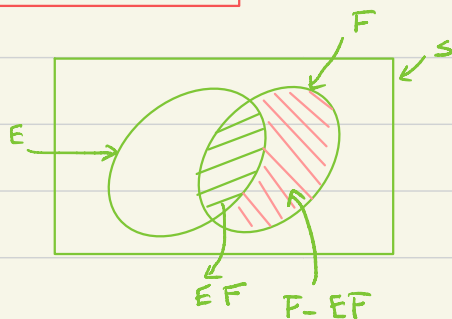
$$P(E \cup F) = P(E) + P(F) - P(EF) \quad (*)$$

Proof:

Partition $E \cup F$ as

$$E \cup F = E \cup (F - EF)$$

disjoint



Hence by Corollary of Axiom 3 (eq. $(*)$ of last lecture):

$$P(E \cup F) = P(E) + P(F - EF)$$



$$= P(E) + [P(F) - P(EF)] \quad \text{by previous Corollary since } EF \subset F$$

$$= P(E) + P(F) - P(EF) \quad \square$$

Corollary

$$P(E \cup F) \leq P(E) + P(F)$$

↑ "Union Bound"

Generalization of $*$ ("Inclusion-Exclusion Principle")

* For 3 events E_1, E_2 and E_3 :

$$P(E_1 \cup E_2 \cup E_3) = P(E_1) + P(E_2) + P(E_3) - P(E_1 E_2) - P(E_1 E_3) - P(E_2 E_3) + P(E_1 E_2 E_3)$$

* For $n \geq 2$ events, $E_1, E_2, \dots, E_n$:

(by induction)

$$P(\bigcup_{i=1}^n E_i) = \sum_{i=1}^n P(E_i) - \sum_{i < j} P(E_i E_j) + \sum_{i < j < k} P(E_i E_j E_k) - \dots + (-1)^{n+1} P(E_1 E_2 \dots E_n)$$

(Probab. of union of n events = sum of probabs of these events taken one at a time, minus the sum of probabs of these events taken two at a time, plus the probabs of these events taken three at a time, and so on.)

Example 1:

An integer between 1 and 100 is chosen at random. What is the probability that it is divisible by either 5 or 7?
(or both)

Answer:

Let E = "chosen integer divisible by 5"
 F = " " " " " by 7"

then

$$P(E \cup F) = P(E) + P(F) - P(EF)$$

$$\text{Here: } P(E) = \frac{|E|}{100} = \frac{20}{100}$$

$$P(F) = \frac{|F|}{100} = \frac{14}{100}$$

$$P(EF) = \frac{|EF|}{100} = \frac{2}{100}$$

since 5 and 7 are relatively prime, and $5|n$ and $7|n$
 $\Leftrightarrow 35|n$

$$\therefore P(E \cup F) = \frac{20 + 14 - 2}{100} = \frac{32}{100} = 0.32.$$

(or 32%)

Example 2:

In a hotel with 300 guests, there are 27 guests who smoke cigarettes, 11 who smoke cigars, 8 who smoke pipes, 4 who smoke both cigarettes and cigars, 3 who smoke both cigarettes and pipes, and 3 who smoke both cigars and pipes. Also there 1 guest who smokes all three. How many smoking guests are staying in the hotel?

Answer:

Let S be set of all guests (sample space) of size $N=300$.

Define events: $E =$ "guests who smoke cigarettes"

$F =$ " " " " cigars"

$G =$ " " " " pipes"

Then $E \cup F \cup G =$ event of guests who smoke.

Want to find: $|E \cup F \cup G|$.

Have, that for any event A in S , $P(A) = \frac{|A|}{|S|} = \frac{|A|}{300}$

Also, know $|E| = 27$, $|F| = 11$, $|G| = 8$

$|E \cap F| = 4$, $|E \cap G| = 3$, $|F \cap G| = 3$, $|E \cap F \cap G| = 1$

∴ By inclusion-exclusion principle,

$$\begin{aligned} \frac{|E \cup F \cup G|}{300} &= P(E \cup F \cup G) = P(E) + P(F) + P(G) - P(E \cap F) - P(E \cap G) - P(F \cap G) + P(E \cap F \cap G) \\ &= \frac{|E| + |F| + |G| - |E \cap F| - |E \cap G| - |F \cap G| + |E \cap F \cap G|}{300} \end{aligned}$$

$$\therefore |E \cup F \cup G| = 27 + 11 + 8 - 4 - 3 - 3 + 1 = \boxed{37}$$

Continuity of the Probability Function

Continuous Function

Recall that a real-valued function $f: \mathbb{R} \rightarrow \mathbb{R}$ is continuous iff for all $x \in \mathbb{R}$ and convergent sequences $\{x_n\}_{n=1}^{\infty}$ with $\lim_{n \rightarrow \infty} x_n = x$, we have that

$$\lim_{n \rightarrow \infty} f(x_n) = f(x)$$

Probability functions have an analogous continuity property, which we explore next.

- We call a sequence of events $E_1, E_2, E_3, \dots$ increasing if
$$E_1 \subset E_2 \subset E_3 \subset \dots \subset E_n \subset E_{n+1} \subset \dots$$

and

decreasing if

$$E_1 \supset E_2 \supset E_3 \supset \dots \supset E_n \supset E_{n+1} \supset \dots$$

• If $\{E_n\}_{n=1}^{\infty}$ is an increasing sequence of events, define

$$\lim_{n \rightarrow \infty} E_n = \bigcup_{n=1}^{\infty} E_n \leftarrow \text{event that at least one } E_n \text{ occurs, } n \geq 1$$

Also if $\{E_n\}_{n=1}^{\infty}$ is a decreasing sequence of events, define

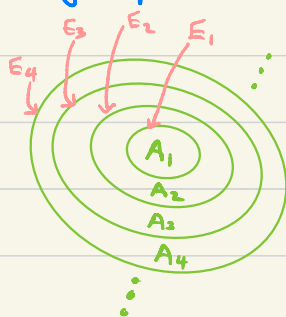
$$\lim_{n \rightarrow \infty} E_n = \bigcap_{n=1}^{\infty} E_n \leftarrow \text{event that every event } E_n \text{ occurs, } n \geq 1.$$

Theorem: If $\{E_n\}_{n=1}^{\infty}$ is an increasing or decreasing sequence of events, then

$$\lim_{n \rightarrow \infty} P(E_n) = P(\lim_{n \rightarrow \infty} E_n)$$

Proof: We will prove the result for an increasing sequence $\{E_n\}$ (a similar proof holds for a decreasing sequence).
(by looking at complements)

$$\text{Set } \begin{cases} A_1 = E_1 \\ A_2 = E_2 - E_1 \\ \vdots \\ A_n = E_n - E_{n-1}, \quad n \geq 2 \\ \vdots \end{cases}$$



By construction, $A_i \cap A_j = \emptyset \quad \forall i \neq j$

$$\text{and } \begin{cases} \bigcup_{i=1}^n A_i \stackrel{(1)}{=} \bigcup_{i=1}^n E_i \stackrel{(1')}{=} E_n \quad \forall n \geq 1 \\ \bigcup_{i=1}^{\infty} A_i \stackrel{(2)}{=} \bigcup_{i=1}^{\infty} E_n \end{cases}$$

Thus

$$P(\lim_{n \rightarrow \infty} E_n) \stackrel{\text{by def}}{=} P\left(\bigcup_{n=1}^{\infty} E_n\right) \stackrel{\text{by (2)}}{=} P\left(\bigcup_{n=1}^{\infty} A_n\right) \stackrel{\text{by axiom 3}}{=} \sum_{i=1}^{\infty} P(A_i)$$

$$= \lim_{n \rightarrow \infty} \sum_{i=1}^n P(A_i) \stackrel{A_i \text{ 's disjoint}}{=} \lim_{n \rightarrow \infty} P\left(\bigcup_{i=1}^n A_i\right)$$

$$\stackrel{\text{by (1)}}{=} \lim_{n \rightarrow \infty} P\left(\bigcup_{i=1}^n E_i\right) \stackrel{\substack{\text{by (1')} \\ = E_n \text{ by (1')}}}{=} \lim_{n \rightarrow \infty} P(E_n) \quad \square$$

Random Selection of Points from Intervals

Won't measure probability of "randomly selecting" a point from a bounded interval $[a, b] = \{x \in \mathbb{R} : a \leq x \leq b\}$. $\leftarrow$ continuous sample space

- For any sub-interval $[\alpha, \beta] \subset [a, b]$, we denote the event "the point falls in $[\alpha, \beta]$ " by $[\alpha, \beta]$. Intuitively for $\alpha < \beta$,

$$P([\alpha, \beta]) = K(\beta - \alpha) \text{ for some } K > 0$$

$\uparrow$
is proportional to length of $[\alpha, \beta]$

Here $S = [a, b]$, thus

$$1 = P(S) = P([a, b]) = K(b - a)$$

$$\therefore K = \frac{1}{b - a}.$$

Thus for $\alpha < \beta$ and $[\alpha, \beta] \subset [a, b]$

$$P([\alpha, \beta]) = \frac{\beta - \alpha}{b - a}$$

Any two sub-intervals in $[a, b]$ that have some length are equally likely to contain the selected point.

• What is the probability of selecting a given point x ?

Consider $a < x < b$ and choose $\varepsilon > 0$ s.t. $[x - \varepsilon, x + \varepsilon] \subset [a, b]$.

Define

$$E_n = [x - \frac{\varepsilon}{n}, x + \frac{\varepsilon}{n}], \quad n \geq 1$$

Then $\{E_n\}$ is a decreasing sequence of events ($E_1 \supset E_2 \supset \dots$)

and

$$\lim_{n \rightarrow \infty} E_n = \bigcap_{n=1}^{\infty} E_n = \{x\}$$

Hence by continuity of P,

$$P(\{x\}) = P(\lim_{n \rightarrow \infty} E_n) = \lim_{n \rightarrow \infty} P(E_n) \stackrel{\text{by } **}{=} \lim_{n \rightarrow \infty} \frac{2\varepsilon/n}{b-a} = 0$$

($\therefore$ selecting pt x from $[a, b]$ is 0 for all x)

• What is the probability of event (α, β) in $[a, b]$?

$$P([\alpha, \beta]) = P(\{\alpha\} \cup (\alpha, \beta) \cup \{\beta\})$$

$$\stackrel{\text{Axiom 3}}{=} \underbrace{P(\{\alpha\})}_{=0} + P((\alpha, \beta)) + \underbrace{P(\{\beta\})}_{=0}$$

$\therefore$

$$P([\alpha, \beta]) = P((\alpha, \beta)) \stackrel{\text{can be chosen similarly}}{=} P([\alpha, \beta]) = P((\alpha, \beta])$$

Notes:

• Fact that $P(\{x\}) = 0$ shows that there are non-empty events with zero probability.

• Since $S = [a, b]$ and $P(S) = 1$, above result implies that
 $1 = P([a, b]) = P((a, b))$

thus there are events that are not equal to S with probability 1.

Example:

A bus arrives at a bus station at random between 8:00 and 8:15 am. Its scheduled arrival time is 8:05 am. We say the bus is "almost punctual" if it is less than 2 minutes early and less than 5 minutes late.

What is the probability that bus is not almost punctual?

Answer:

Here $S = [8:00, 8:15]$,
length = 15 minutes

length is 7 minutes

$E = [8:03, 8:10] = \text{"bus almost punctual"}$

Want find $P(E^c)$:

$$P(E^c) = 1 - P(E)$$

$$= 1 - \frac{7}{15} = \boxed{\frac{8}{15}}$$